

Class #8 - Linear Models

1. One serving (28 g) of Kellogg's Cracklin' Oat Bran supplies 110 calories, 3 g of protein, 21 g of carbohydrate, and 3 g of fat. One serving of Kellogg's Crispix supplies 110 calories, 2 g of protein, 25 g of carbohydrate, and 0.4 g of fat.
 - (a) Set up a matrix B and a vector u such that Bu gives the amount of calories, protein, carbohydrate, and fat contained in a mixture of three servings of Cracklin' Oat Bran and two servings of Crispix.
 - (b) Suppose that you want a cereal with more protein than Crispix but less fat than Cracklin' Oat Bran. Is it possible for a mixture of the two cereals to supply 110 calories, 2.25 g of protein, 24 g of carbohydrate, and 1 g of fat? If so, what is the mixture?

2. In a certain region, about 4% of the city's population moves to the surrounding suburbs each year and about 3% of the suburban population moves into the city. In 2026, there were 600,000 residents in the city and 400,000 in the suburbs. Set up a difference equation that describes this situation, where x_0 is the initial population in 2026. Then estimate the population in the city and in the suburb two years later, in 2028. (Ignore other factors that might influence the population size.)